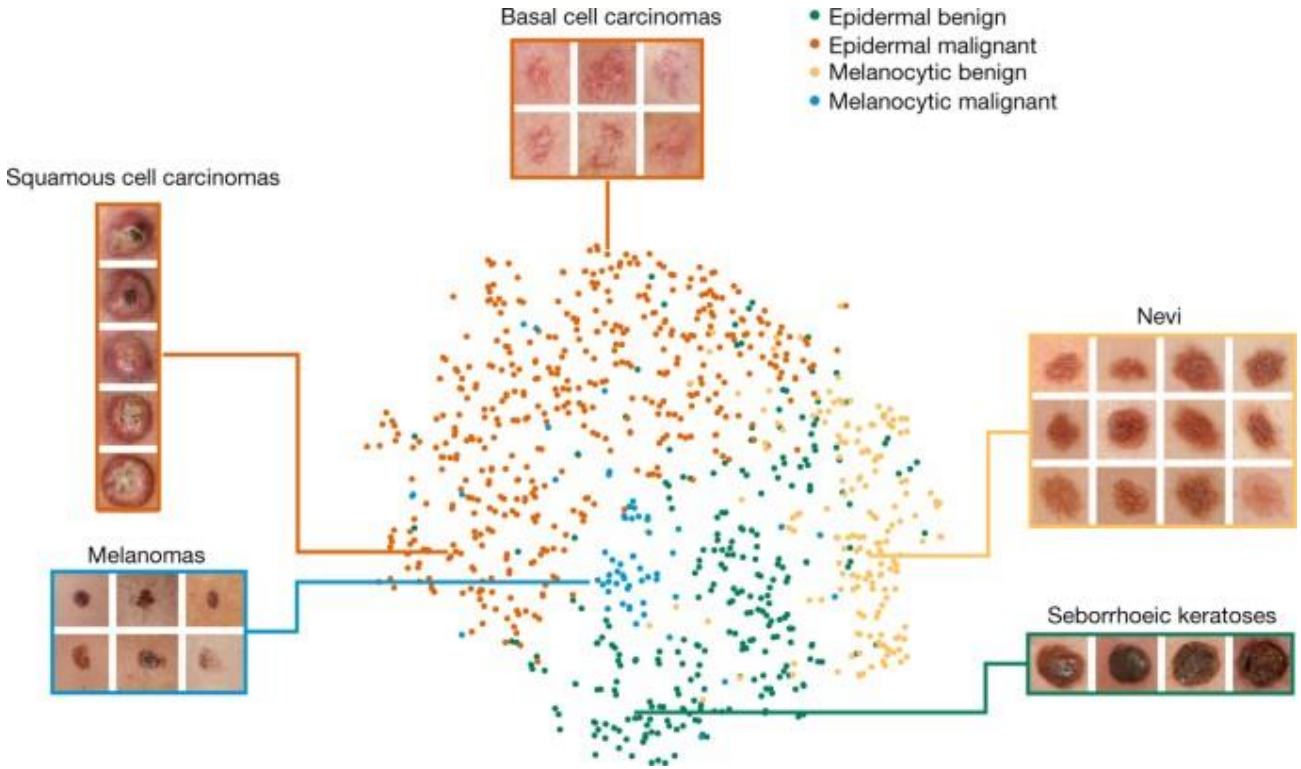
04

Advanced Deep Learning Topics

Transfer learning



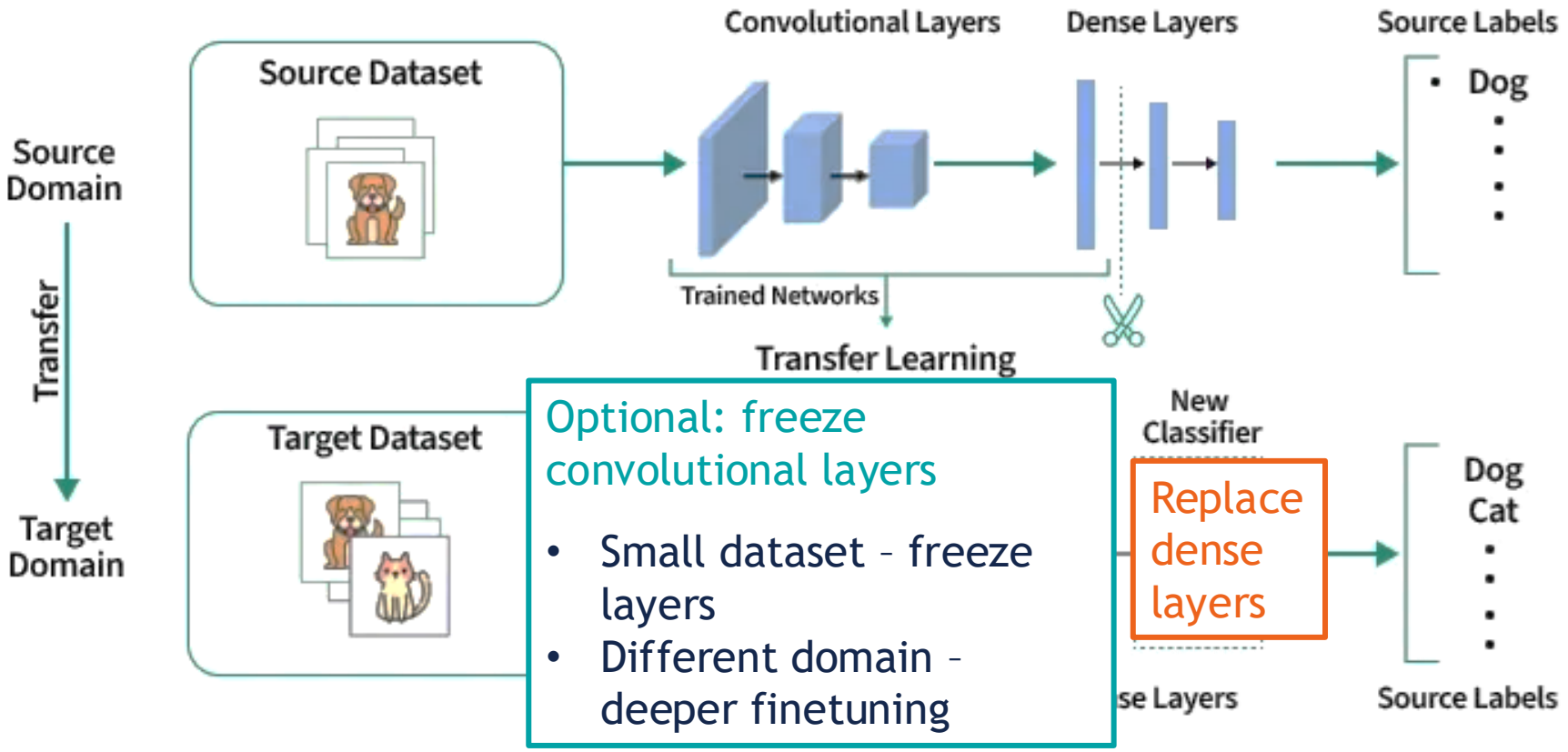
Deep learning is all about scaling →

Where do you find millions of labeled carcinoma images?

Thrun et al., Dermatologist-level classification of skin cancer with deep neural networks, Nature (2017)

Transfer learning

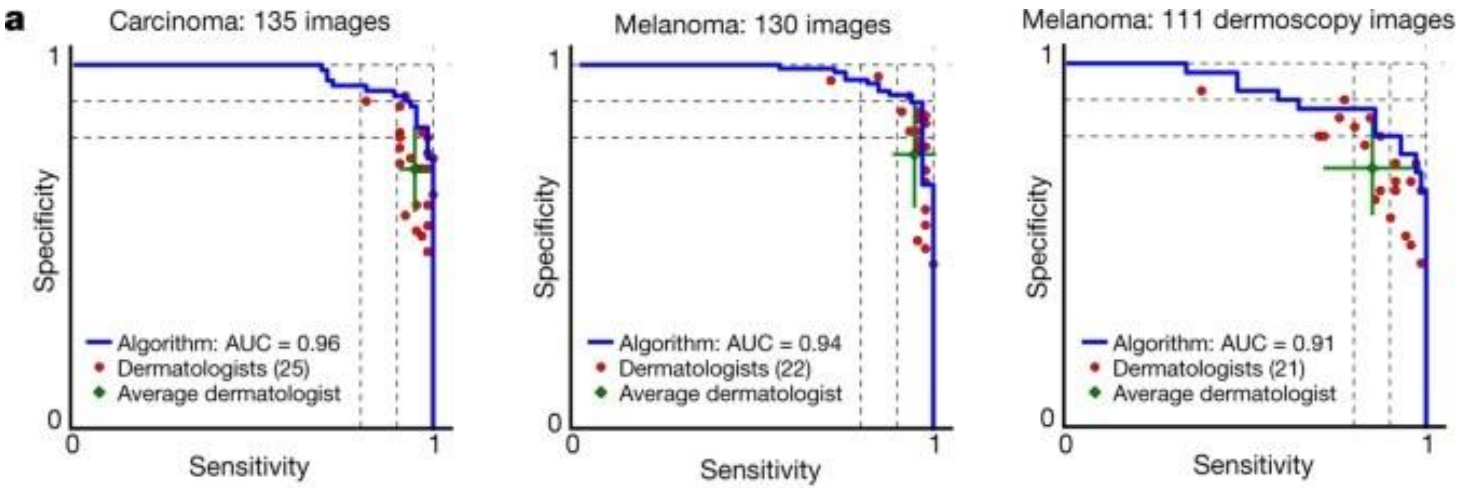
taking a *pre-trained model*, and *finetuning* it on a domain-specific dataset



https://www.geeksforgeeks.org/machine-learning/ml-introduction-to-transfer-learning/

Transfer learning

- Pretrained model (trained on ImageNet), downloaded from Github
- Full finetuning on ~150’000 carcinoma images



Thrun et al., Dermatologist-level classification of skin cancer with deep neural networks, Nature (2017)

Explainability

Interpretability
(how the model works)

Input → BLACK BOX → Prediction

Explainability
(why the model made a specific prediction)

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Explainability

Interpretability

- Simple models
- Hybrid models
  (encoder + simple model)
- Domain-driven architecture
  (e.g. PASNet)

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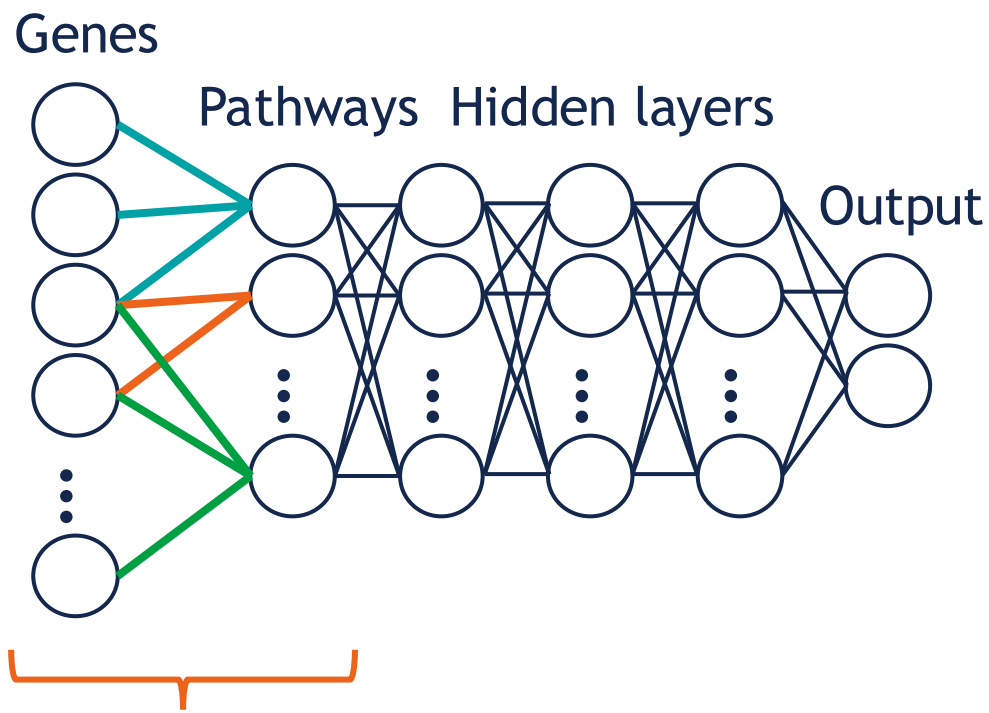

Enforce domain knowledge
(known pathways)

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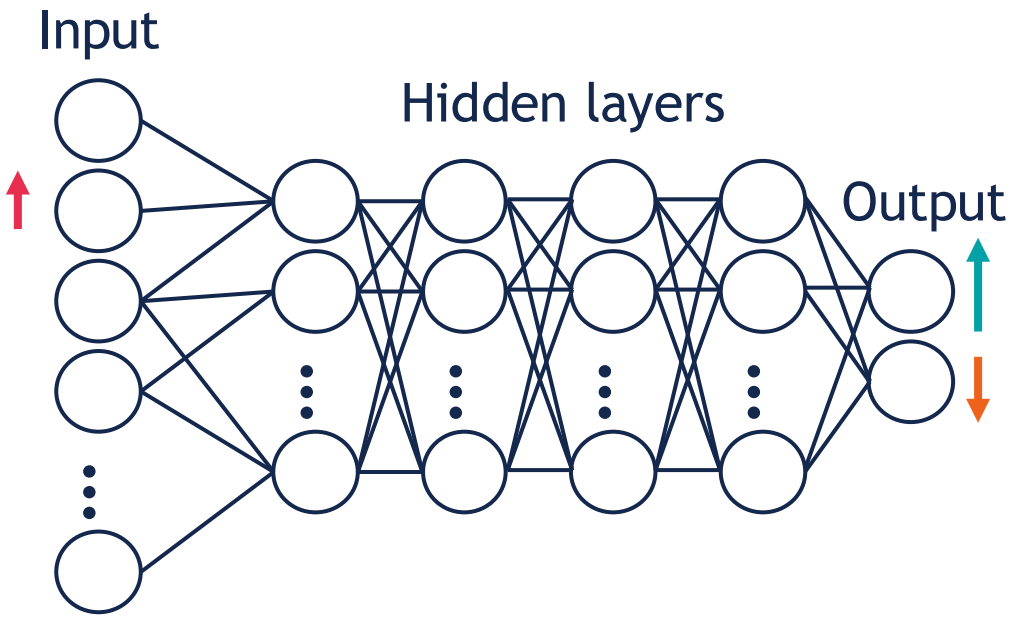
Explainability

- Saliency maps
- Perturbation
- SHAP
- Feature visualization
- Attention analysis



If I change this input slightly, how does it affect the output?

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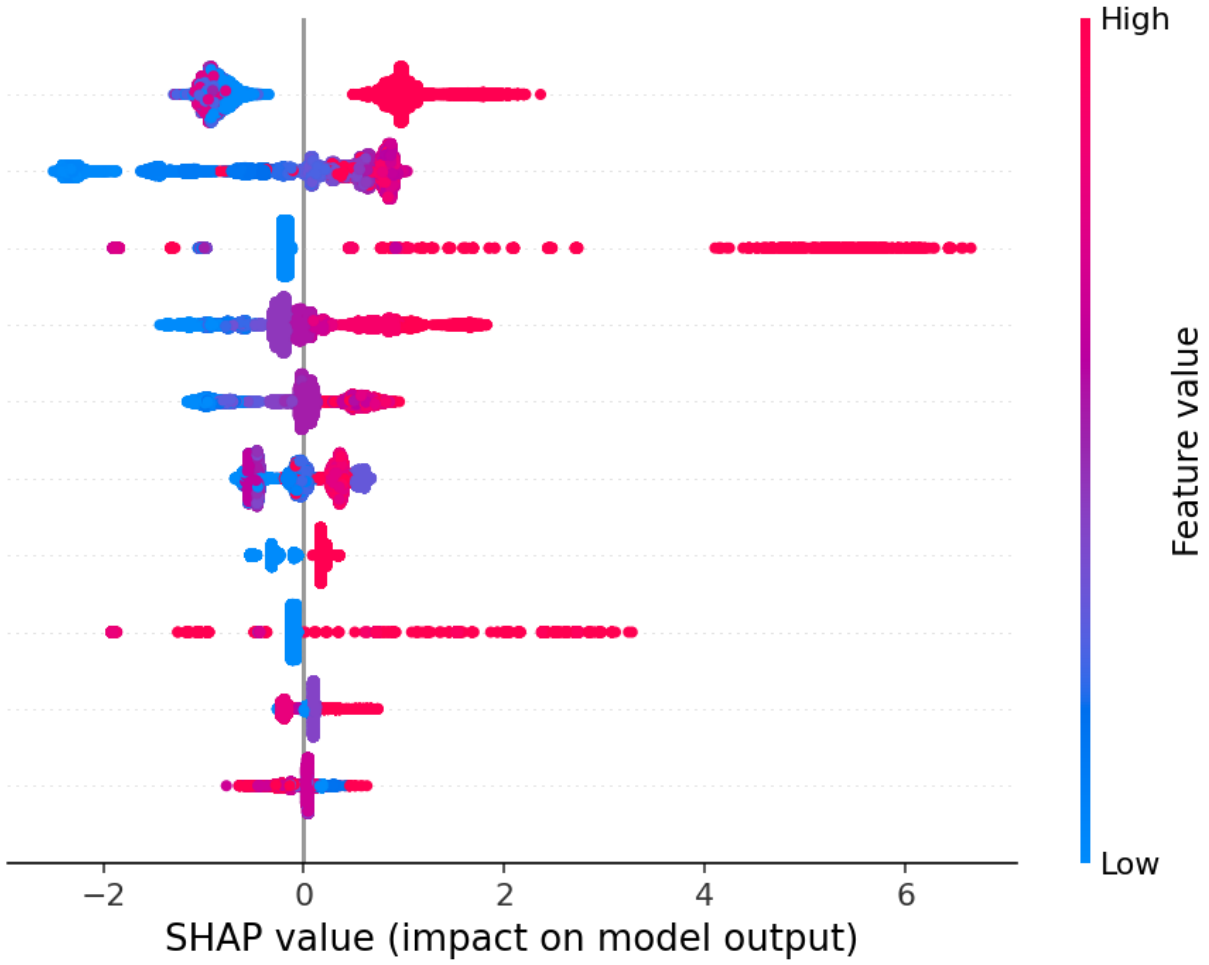

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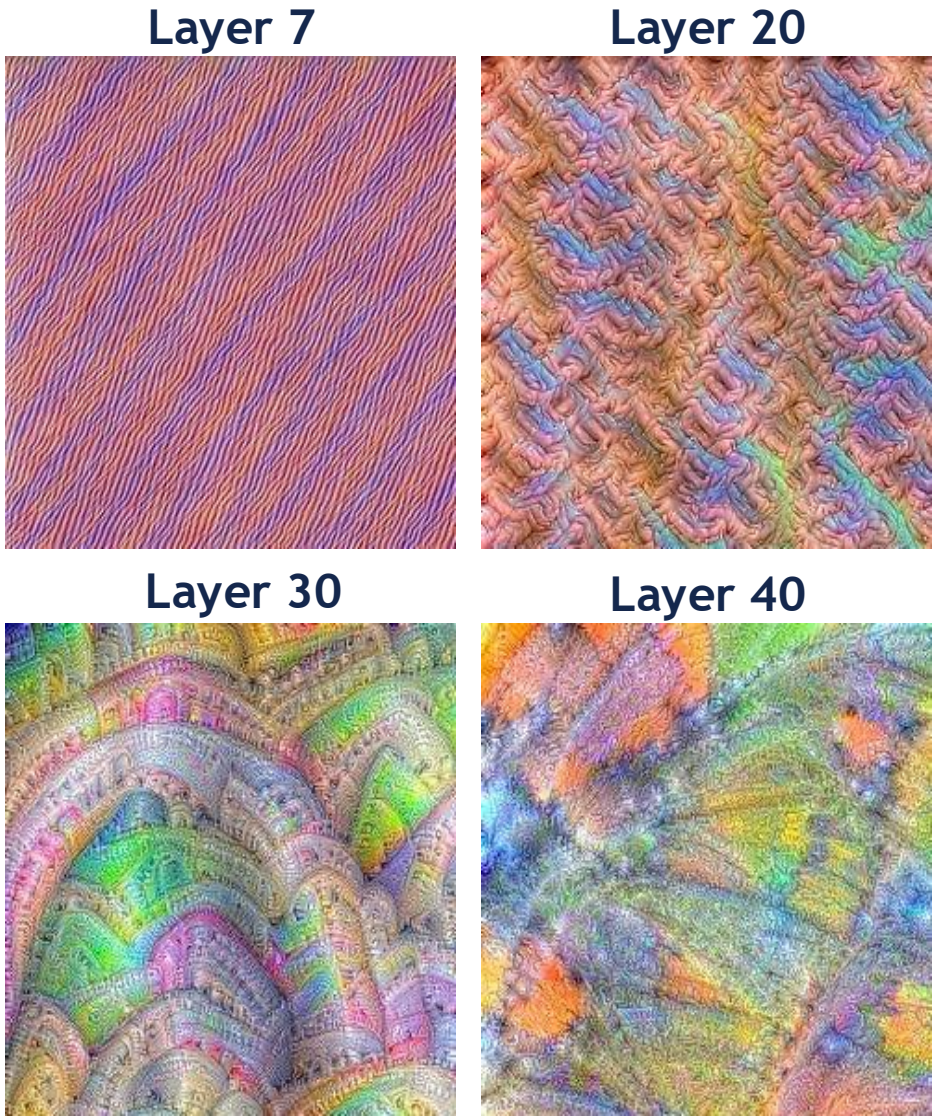

Which input maximizes the activation of a specific node?

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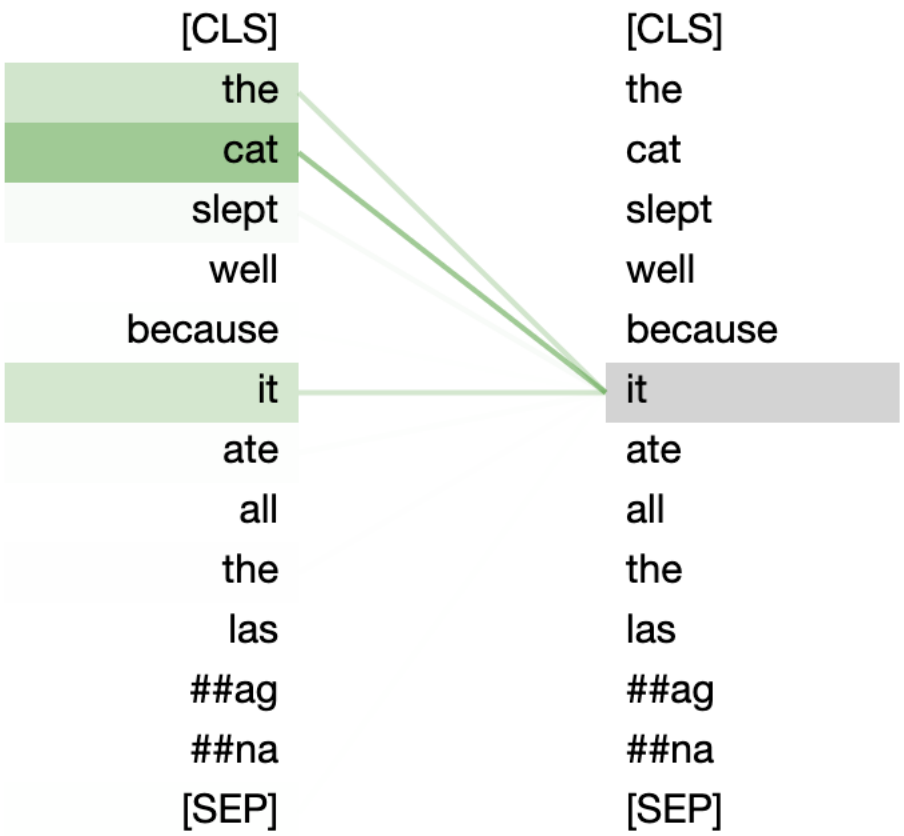

*Attention shows where the information flows, **not how it is being used***

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Correlation

≠

Causation

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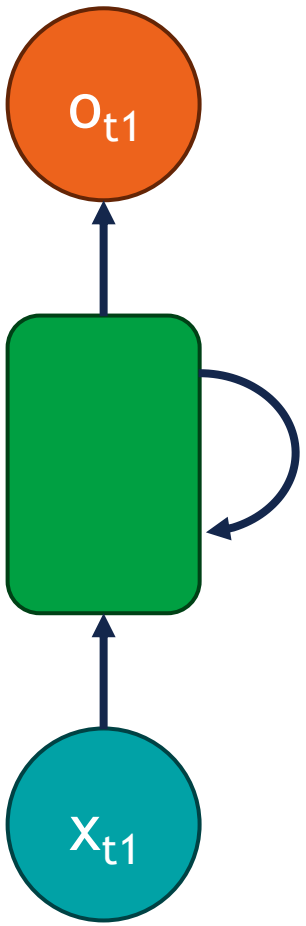
Recurrent Neural Networks

- CNN: spatial information
  RNN: sequential information
- Hidden state over time (~memory of the previous inputs)
- Step-by-step processing (slow!)
- Vanishing gradients (information from previous steps fades away)
- *LSTM/GRU: adaptations for better long-term memory*



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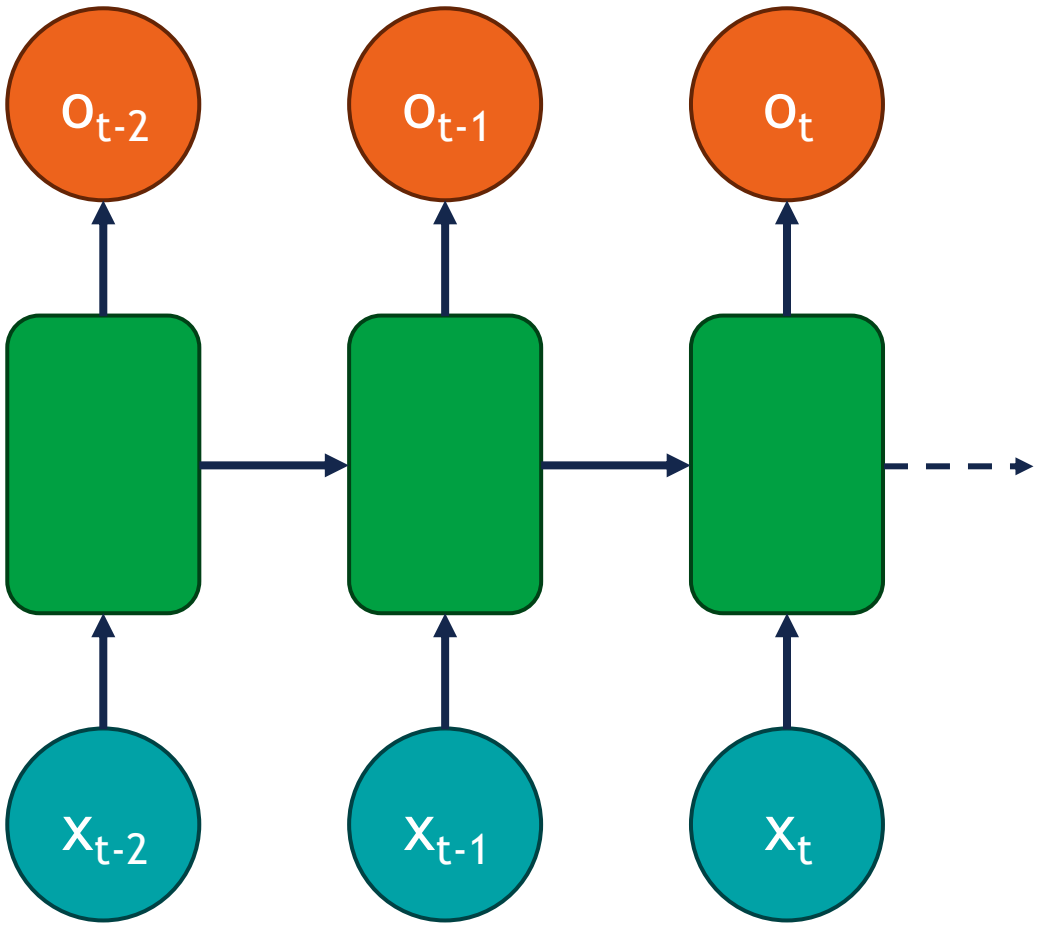
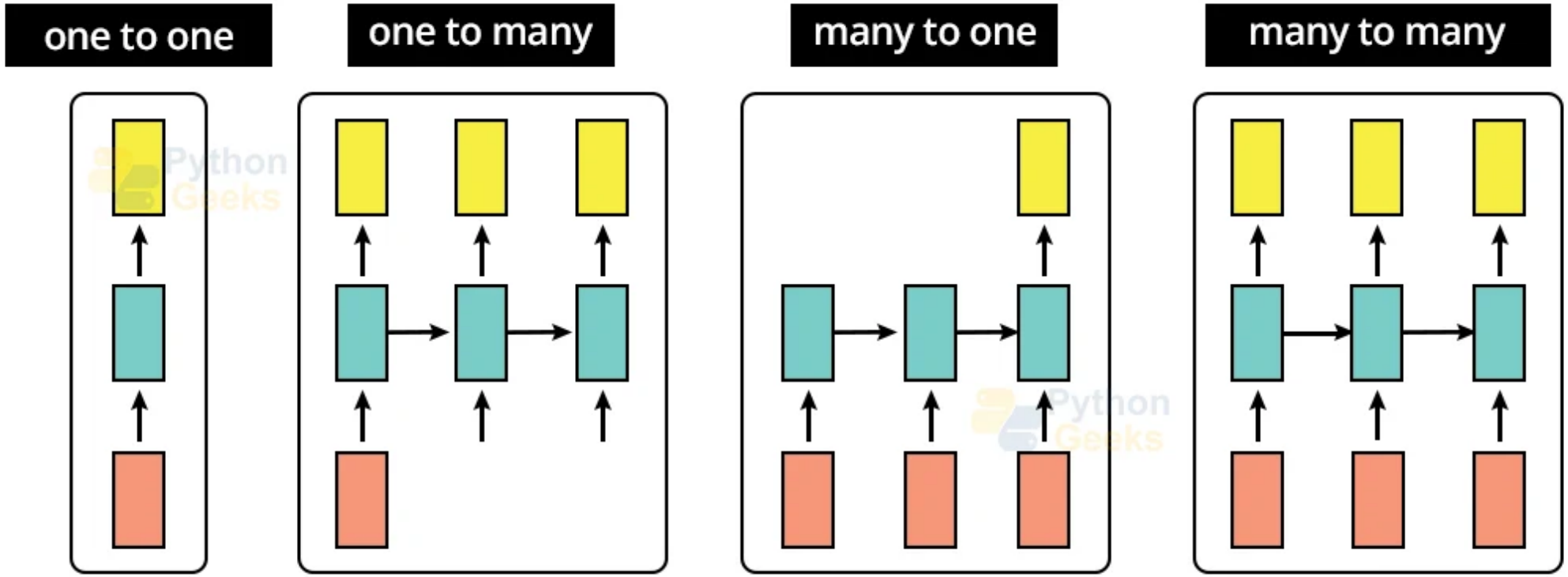


Recurrent Neural Networks



Vector → sequence
protein sequence generation from a feature vector

Sequence → label
DNA sequence classification

Sequence → sequence
RNA secondary structure labeling

ChatGPT

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GPT - General Pretrained Transformer

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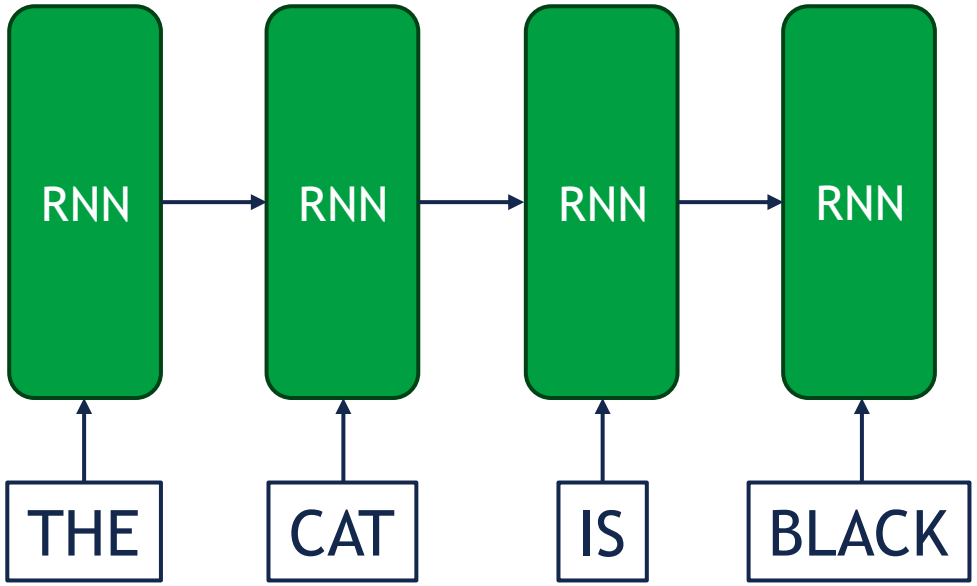
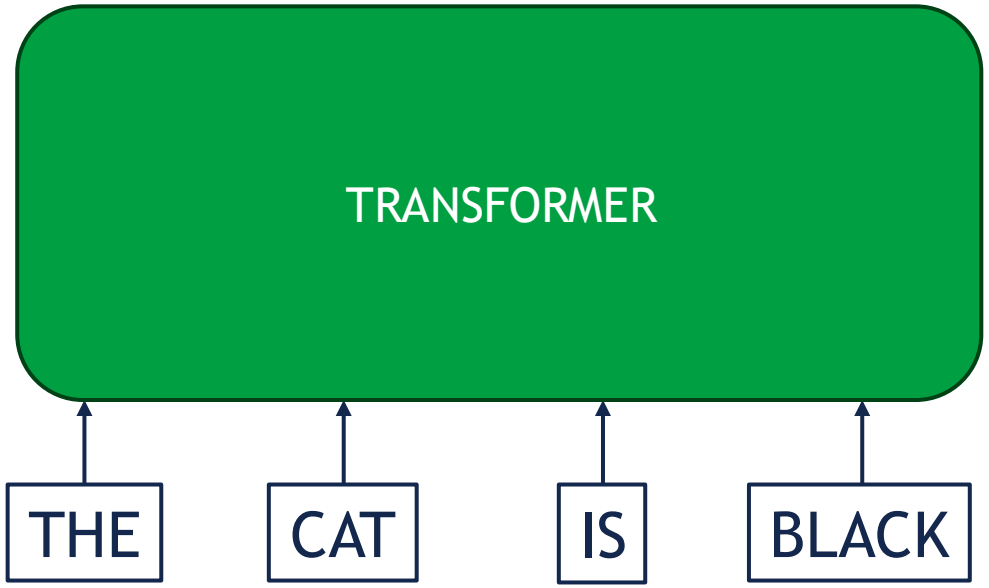
Transformers

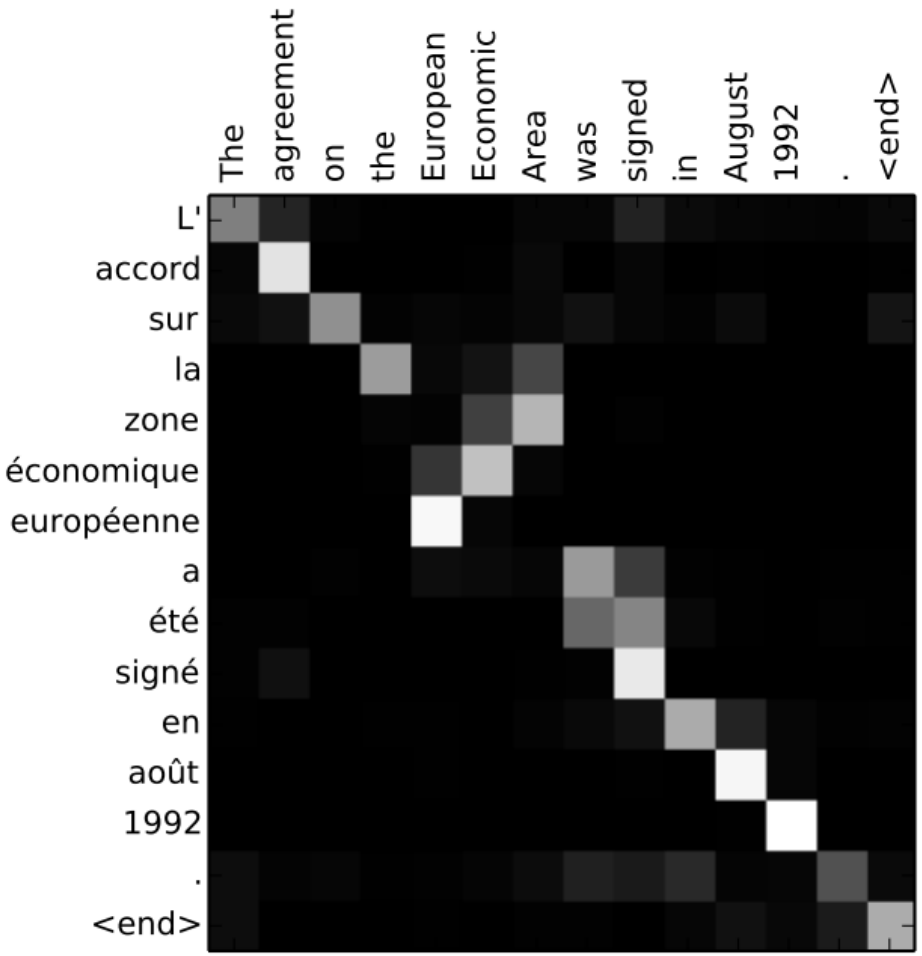


Input is processed sequentially (step-by-step)



Input is processed in parallel (all-in-one)

Transformers



- **Self-attention mechanism** learns to decide which parts are most important
- State-of-the-art models (ChatGPT, ViT, etc.)

Self-supervised learning

Transformer models require huge amounts of data, but labeled data is unavailable or expensive → generate your own labels

Autoencoding

- Reconstruct the original input from an altered sample (e.g. denoising)

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Masked modeling

- Mask parts of the input
- Predict the full (original) sample

"The cat is [redacted] *on the mat"*

↑

Predict the masked word

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Self-supervised learning

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Autoencoding

- Reconstruct the original input from an altered sample (e.g. denoising)

Masked modeling

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- Predict the full (original) sample

Positive pair

"The cat is sitting on the mat"
"The cat sits on the mat"
"I like pizza"

Negative pair

Contrastive learning

- Based on augmentations (~variations of a sample)
- Push similar samples closer
- Push dissimilar ones apart

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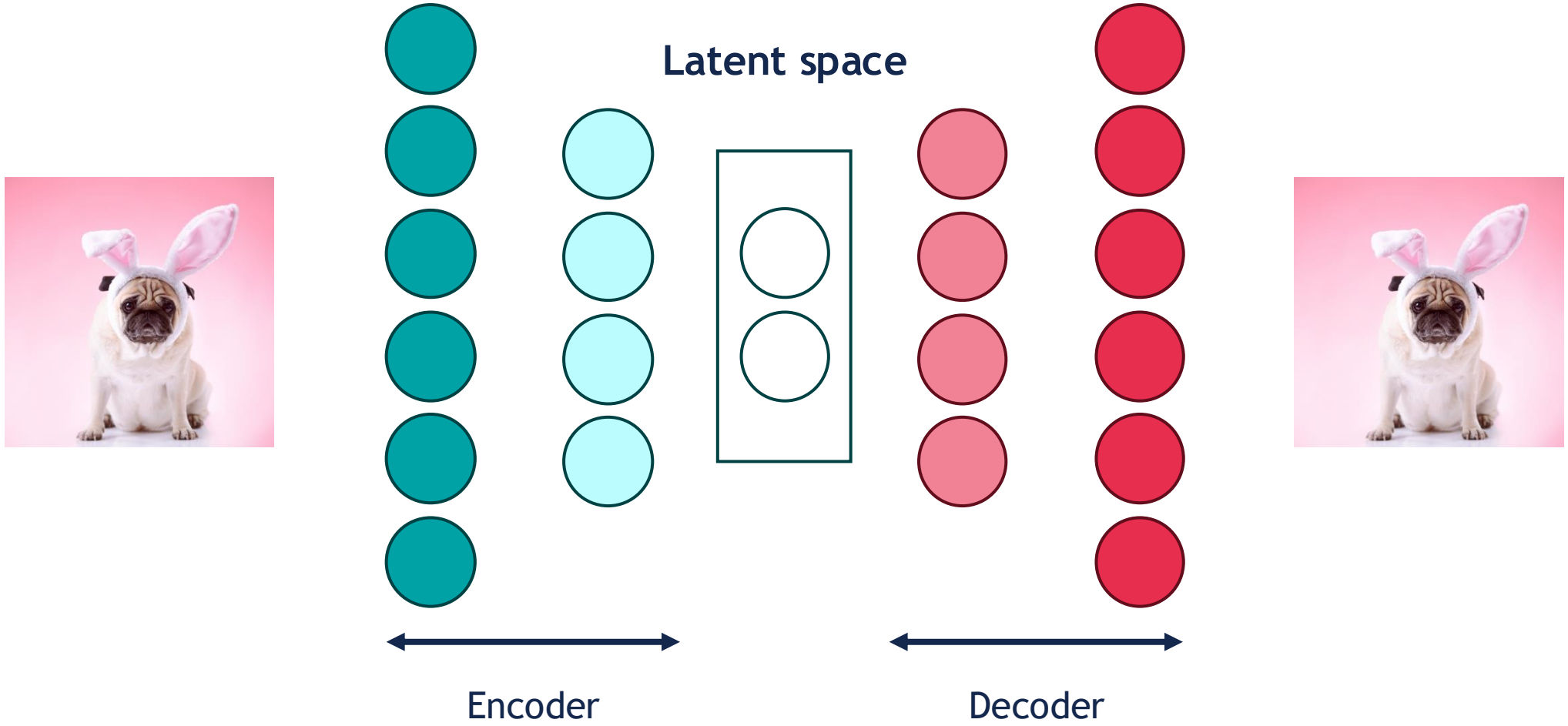
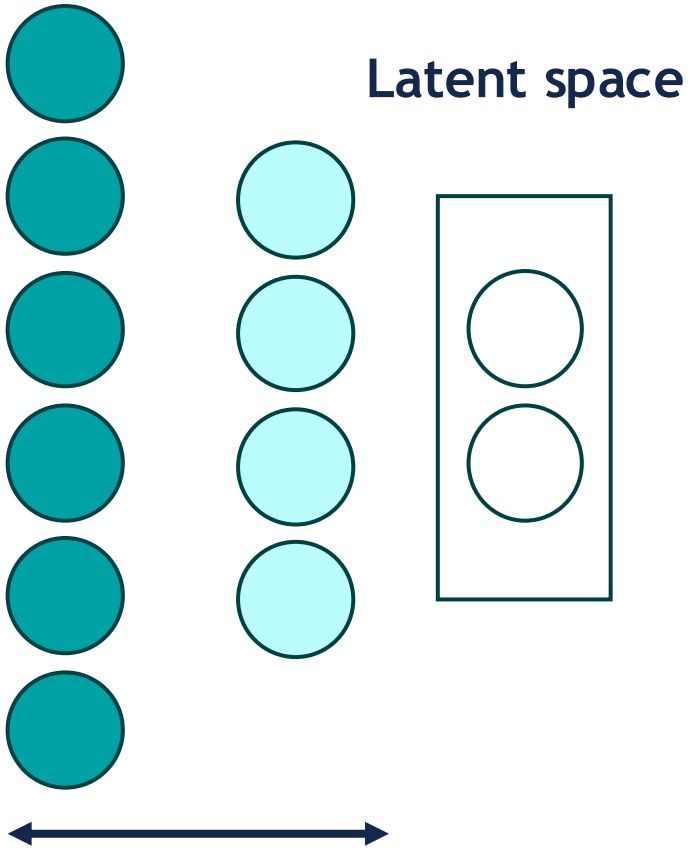
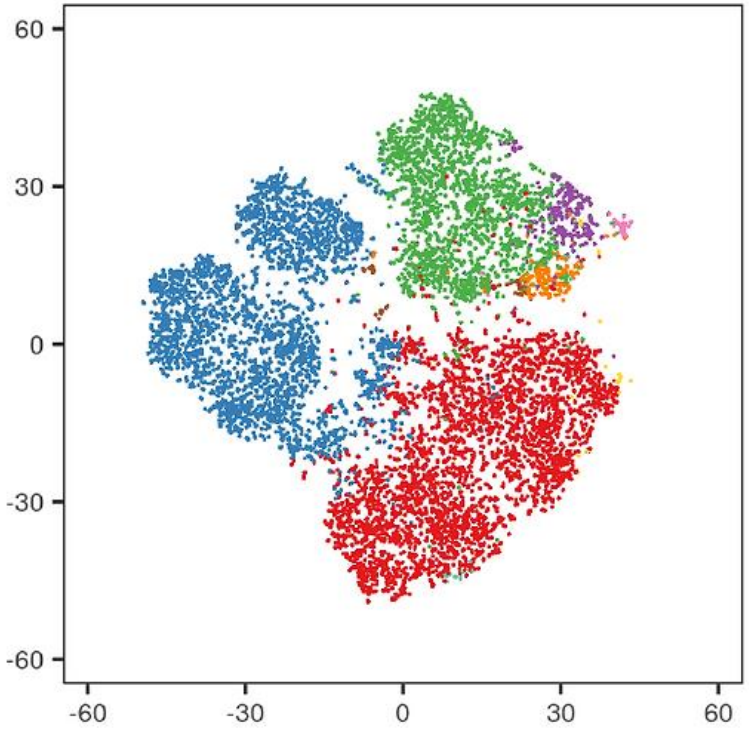
Autoencoder



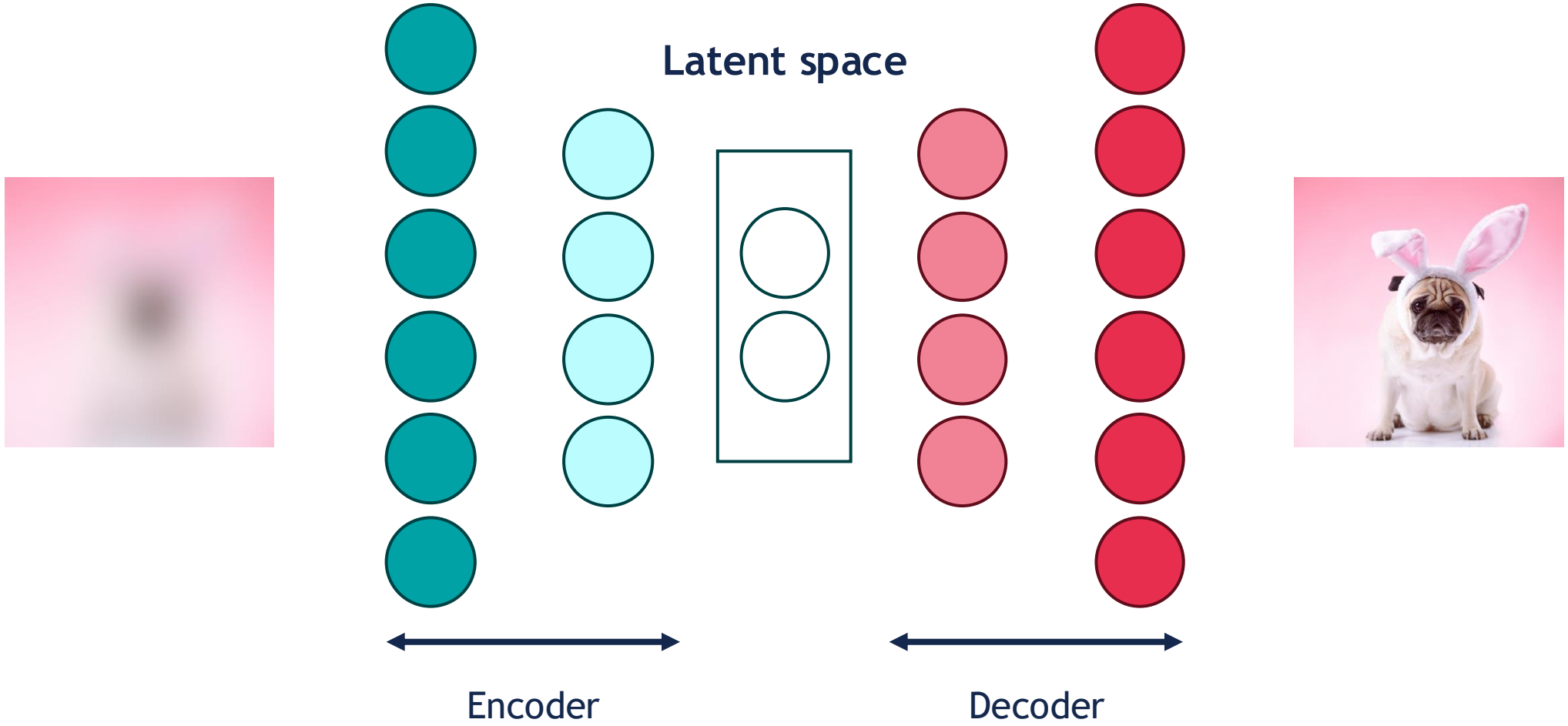
Autoencoder - dimensionality reduction



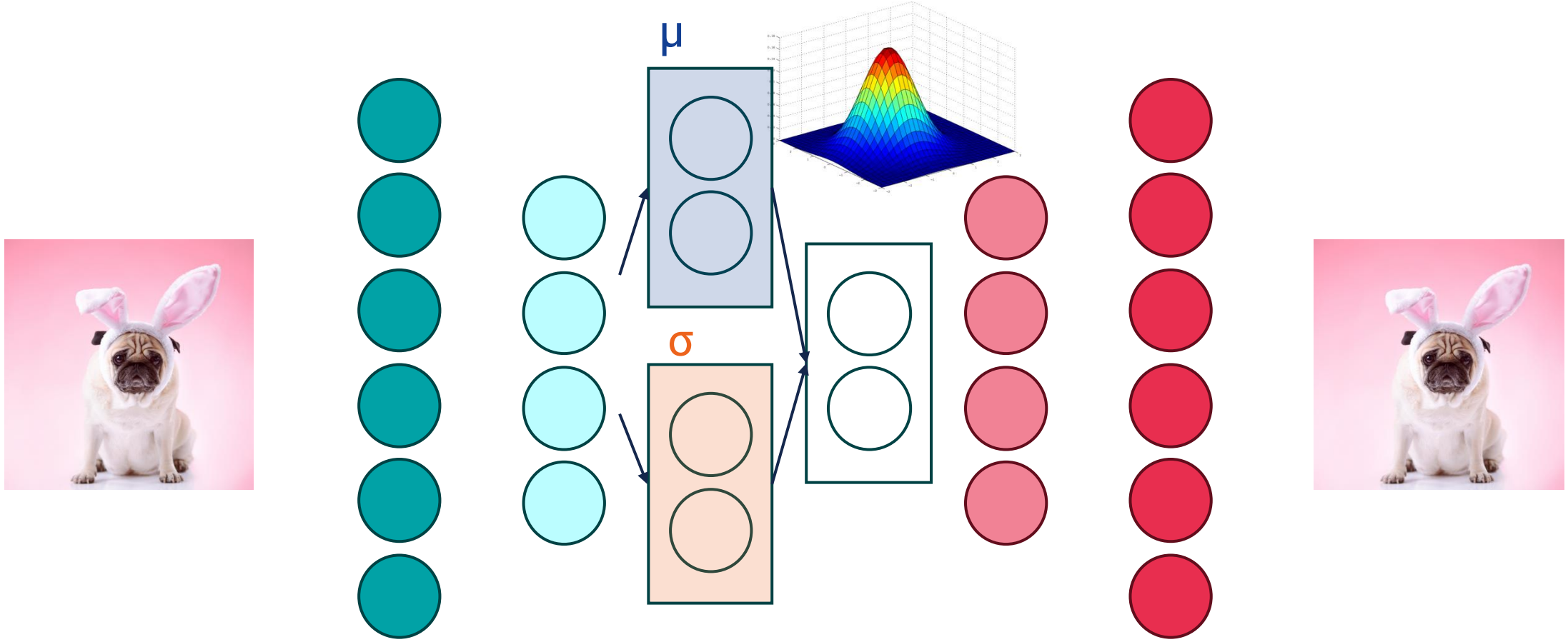
Low-dimensional feature vector



Autoencoder - denoising

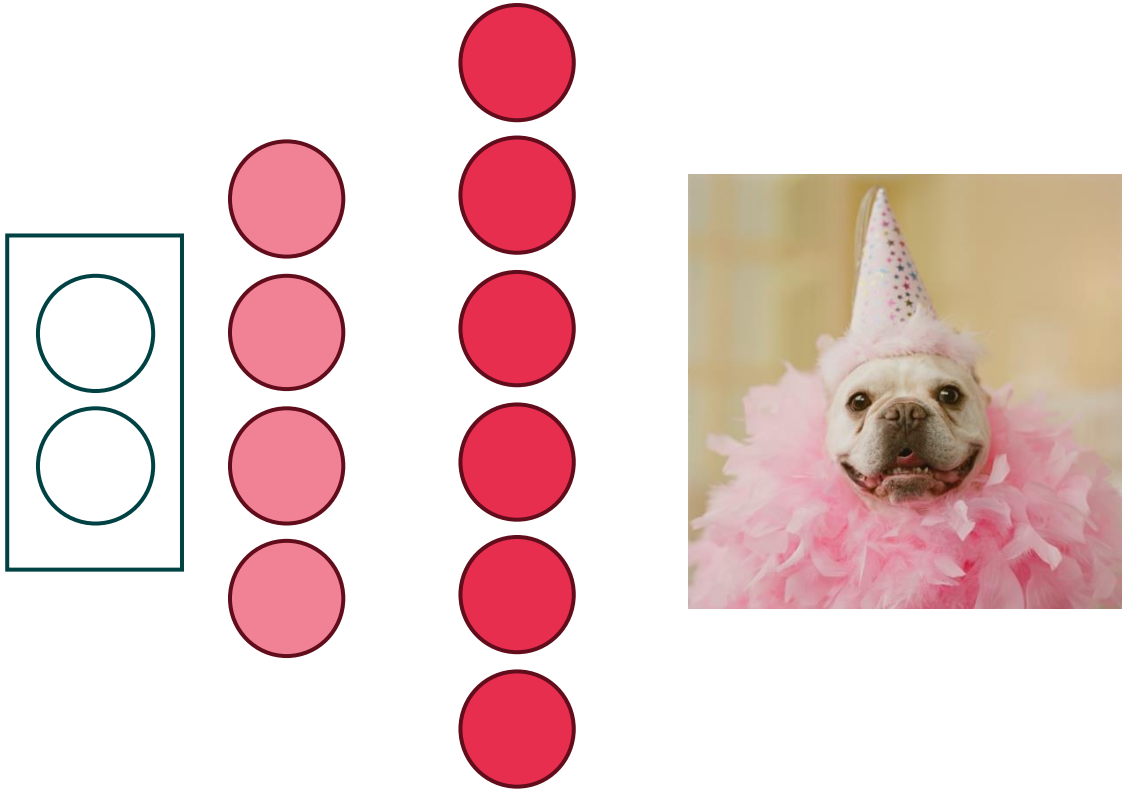


Variational autoencoder - generative AI

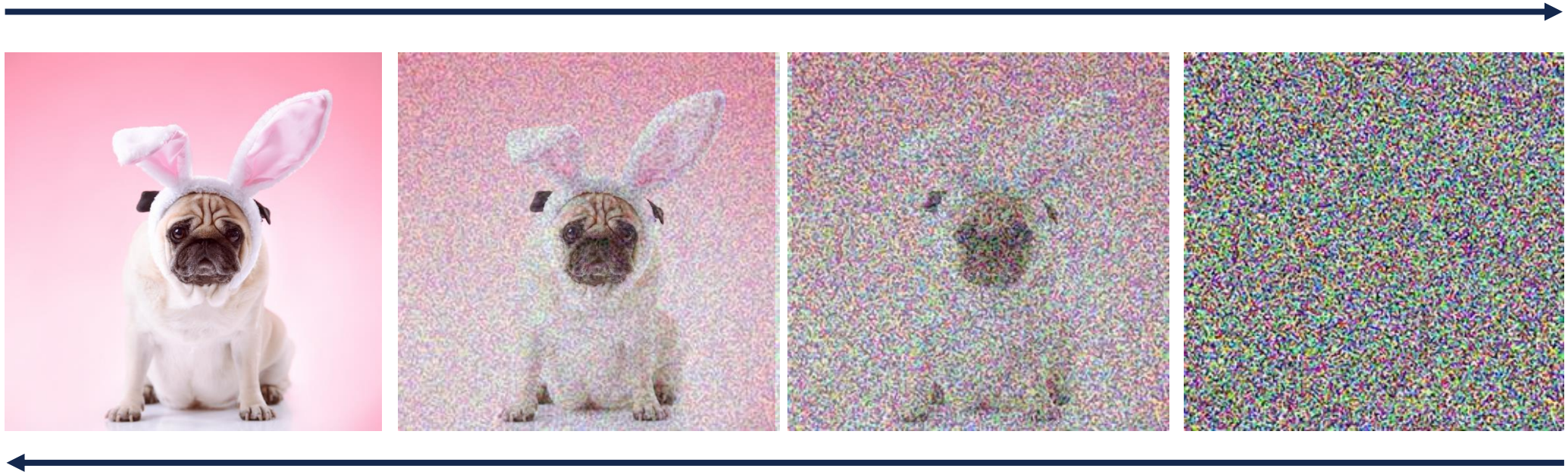


Picking a random latent variable can generate garbage output
→ map input to a distribution in the latent space (μ and σ)

Variational autoencoder - generative AI

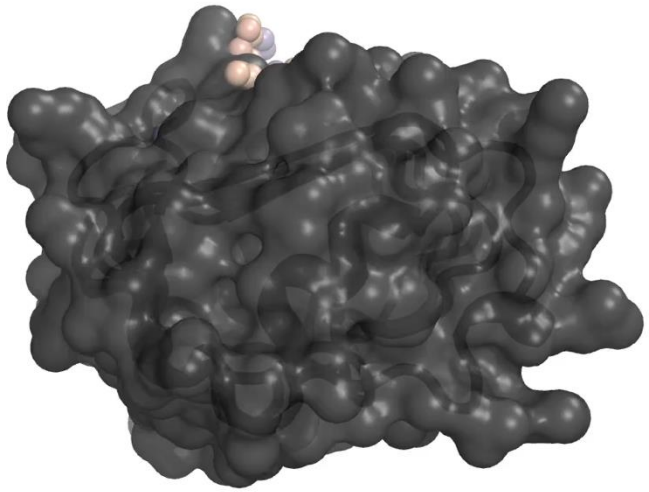


Diffusion model



Diffusion models learn to denoise images blurred with noise (i.e. reversing the diffusion process)

Diffusion model



RF Diffusion – uses guided diffusion to generate brand-new *protein structure* candidates that satisfy design constraints

https://www.bakerlab.org/2023/03/30/rf-diffusion-now-free-and-open-source/

Jolan Heyse

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